

Classical Mechanics(H1) (SC1.102)
IIIT-H, Semester Winter 26, Assignment 1

Submission deadline: 1st February 2026

1. For the following systems, find the *a)* degrees of freedom, *b)* chose a set of suitable generalized coordinates, *c)* write the Lagrangian, and *d)* write down the equation of motion for each coordinates (you are not required to solve the equations)
 - (a) A double pendulum: the first pendulum is of length ℓ_1 and mass m_1 and is suspended from the ceiling. The second pendulum is of length ℓ_2 and mass m_2 and is suspended from m_1 .
 - (b) A simple pendulum where the pivot moves along a horizontal X -axis as $x(t) = \sin(\omega t)$.
 - (c) A simple pendulum where the pivot moves along a vertical axis.
 - (d) See figure 1. There is a mass m_2 that moves on a vertical axis. The entire system rotates about the axis with constant angular velocity Ω .
 - (e) Two identical pendulums of mass m and length ℓ are suspended from a ceiling at a distance d . The two mass points are connected to each other by a massless spring of length d (when in relaxed) and spring constant k .
 - (f) Figure 1 of lecture note.

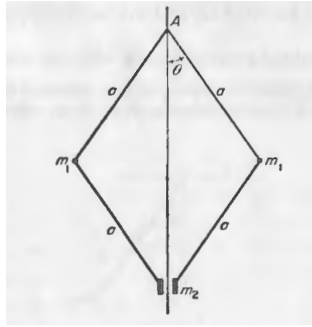


Figure 1:

2. Consider a system described a set of generalized coordinates $q(t) \equiv \{q_1(t), q_2(t), q_3(t), \dots, q_f(t)\}$. Starting from the Hamilton's principle show that each of the generalized coordinates satisfy

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

Hint: Your answer should reflect the your understanding.

3. Consider a system of N particles described by a set f generalized coordinates $q \equiv q_1, q_2, \dots, q_f$. If $\mathbf{r}_i$ denotes the position vector and m_i is the mass of the i^{th} particle then the generalized kinetic energy is

$$T = \sum_j^f \sum_k^f a_{jk}(q) \dot{q}_j \dot{q}_k, \quad (1)$$

where

$$a_{jk} = \frac{1}{2} \sum_{i=1}^N m_i \left(\frac{\partial \mathbf{r}_i}{\partial q_j} \cdot \frac{\partial \mathbf{r}_i}{\partial q_k} \right).$$

4. Consider that the cost of flying an aircraft at height (vertically up) z is $e^{-\kappa z}$ per unit distance of flight path, where κ is a positive constant. Assume that the aircraft flies in the $x - z$ plane. Find the differential equation for z so that the flying cost will be minimum. You don't have to solve the equation. *Hint: Use calculus of variation.*
5. Consider a Lagrangian $L(q, \dot{q}, t)$. A new Lagrangian is formed such that $L'(q, \dot{q}, t) = L + \frac{dF(q, t)}{dt}$. Starting from the EL equation of motion show that the EL EoM for L' remain unchanged. *Hint: perform partial derivative carefully.*